Farhan Mahmood

Full-Stack Software Engineer | Java | Spring Boot | React.js | Next.js | Python | AI/LLM Integration

farhanmahmood2804@gmail.com | +91-7004048940 | linkedin.com/in/10-farhan-mahmood | farhan2804portfolio.netlify.app

geeksforgeeks.org/profile/el3317farh8k4v | leetcode.com/u/farhan28

EXPERIENCE

Software Development Intern | Sikharthy Infotech Pvt. Ltd. | Jun 2023 – Aug 2023
- Developed web application features using **JavaScript and React.js** within a 15-person development team.
- Built and integrated frontend functionality with **server-side application logic and REST-based workflows**.
- Collaborated with team members to debug issues, improve features and maintain application functionality.

Full Stack Development Trainee | Kodnest | Jan 2025 – Jun 2025
- Completed a six-month full-stack development program focused on **Java, Python, JavaScript, SQL and software testing**.
- Built hands-on applications using **Java, Spring Boot, REST APIs and relational databases**.
- Practiced **OOP, Data Structures & Algorithms, API integration, debugging and software testing**.

PROJECTS

Diagnostic Test Booking & Payment Management System
Java 21 | Spring Boot | Spring Security | JWT | PostgreSQL | Spring Data JPA | REST APIs | Swagger/OpenAPI | JUnit/Mockito
https://github.com/farhan2804/eve_healthcare_booking_api
- Developed a backend healthcare platform supporting **authentication, diagnostic test management, appointment booking and payment workflows**.
- Designed REST APIs and relational database schemas for users, diagnostic centres, tests, bookings and payments.
- Implemented request validation, authorization, payment ownership checks, error handling and **idempotent payment webhooks**.
- Added Swagger/OpenAPI documentation, structured logging, JUnit/Mockito unit tests and integration tests, with **9 automated tests passing**.

Real-Time Predictive Maintenance Platform
Python | FastAPI | scikit-learn | Java | Spring Boot | React.js | PostgreSQL
https://github.com/farhan2804/predictive_maintenance
- Developed a multi-service platform for monitoring industrial equipment using temperature, vibration and pressure sensor data.
- Built **Spring Boot REST APIs** and integrated a Python/FastAPI prediction service for equipment failure prediction.
- Developed a React dashboard for displaying sensor readings, equipment status and prediction results.
- Connected the frontend, Java backend, Python ML service and PostgreSQL database in a multi-service architecture.

AI SQL Assistant – Natural Language to SQL Engine
Python/LLM Integration | Groq Llama 3.3 | Java | Spring Boot | React.js | PostgreSQL | REST APIs
https://aisqlassistant.netlify.app/
- Developed a full-stack application that converts **natural-language requests into SQL queries** using an LLM API.
- Built Spring Boot REST APIs for SQL generation and PostgreSQL database interactions.
- Developed React components for query input, API communication and displaying generated SQL and results.
- Deployed the application using **Netlify, Railway and Neon PostgreSQL**.

SKILLS

Languages: Java 21, Python, JavaScript, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Material UI

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, JWT Authentication

Databases: PostgreSQL, MySQL, Neon PostgreSQL

AI & LLM: LLM Integration, Generative AI, FastAPI, Scikit-learn, Random Forest

VLSI & Electronics: VLSI Design Flow, Digital Electronics, Verilog/VHDL fundamentals, FPGA fundamentals

Tools & Platforms: Git, GitHub, Postman, Swagger/OpenAPI, Azure AI, Vercel, Railway

EDUCATION

Bachelor of Technology (B.Tech) Aug 2021 – Jun 2024
Electronics and Communication Engineering

University: Jharkhand University of Technology
CGPA: 8.04

Diploma
Electrical Engineering

University: Jharkhand University of Technology
CGPA: 8.21

CERTIFICATIONS

Microsoft Certified: Azure Cloud Fundamentals (AZ-900)

Microsoft Certified: Azure AI Fundamentals (AI-900)

ADDITIONAL — TECHNICAL EDUCATOR, YOUTUBE

Engineering & Web Development Channel
https://www.youtube.com/@formulaC_Coder2804